

- **Title: social network analysis on proteins interactions in the human body**

- **Aim: to use the centrality analysis on proteins**

The aim of this analysis is to see which proteins are the most important in the space (in the human body) and their role in fighting viruses and to study the interaction between them and visualize which one or group of nodes is more important in the proteins interaction network

- **Hypothesis: we will use the node centrality analysis to see all four types of importance, them being the degree centrality and the betweenness and closeness, and lastly the eigenvector which is the hardest one to get because it needs a bit more calculations and the project will go as follows:**

- **Materials: the tools we used for this project are cytoscape, R studio, excel**

For clarification: cytoscape can take care of showing the degree and betweenness and closeness but as mentioned before the eigenvector will need more steps and we will use R for that

Cytoscape: to visualize and to have the option if taking a subset when needed, and also for the ease of use since it has an interface

R Studio: we will use R to make sure our calculations are correct and accurate and above all be able to get all the centrality attributes in case Cytoscape fails to do so

As for excel: it's just to view the files in a way everyone can read them and understand them

And with that we dive onto how we used our tools to analyze this dataset

- Procedure:

A. what will the programs do:

1. Cytoscape to open and visualize the XML file for future use, and to be able to make a subset of the network which in our case here is very massive

2. In R Studio we used some outputs from Cytoscape to make all the calculations needed and how to get the final output

3. Excel will just show us the results/output and gives a way to interact with them

B. what the input of the program will be?

1. The XML File (the dataset itself)

2. The code that will convert the dataset file into readable Excel files

C. what the output of the programs will be?

1. Describe the data

File name:

BIOGRID-ORGANISM-Homo_sapiens-3.5.174.psi25.xml

Our data set contains biological interaction data specific to the organism Homo sapiens (human) and is formatted in the PSI-MI 2.5 standard for protein-protein interaction data exchange, and was made by BioGRID

2. Our project steps and methods (with screenshots)

We imported the XML file to Cytoscape as a network and it looks as follows

Admittedly the dataset is very massive and our computers can't handle that level of calculations, so we will be taking a subset (even twice) to get a data we can work with we will be using the plugin "CytoNCA" to give us a frame of reference and will do so as follows

After we are satisfied with our new subnetwork we will extract the "edge table" into a CSV file and save it somewhere R can access it (we will store it in documents) as follows

For now other than visualizing our dataset we are pretty much don't with Cytoscape and ready to move onto R studio

**Using the CSV file and since we got here let's start
Discussing the Code**

- Code Discussion:

Since at the time we didn't know we could just plainly export the CSV file we had to take some "Extreme measures" it being using the following libraries **screenshot**

Using Rcy3 package we were able to let R use and control Cytoscape as long as its open on the subnetwork view, we did that and were able to extract all the data we need from the dataset opened in Cytoscape as follows: **screenshot**

Then we will need to do some data cleaning in a way that R can easily get us the results we want using them it will be split into two parts one for the nodes and one for the edge

1. Nodes code **screenshot**
2. Edges code **screenshot**

After the cleaning is done we then use the (igraph) library to import both data frames to make the g or (graph inside a variable g and g2) as well as getting 3 of the four attributes we wanted **screenshot**

We have one last attribute that we need to get which is the igenvector we think using the provided command gives false results so we decided to do it by ourselves we did that by

making the adjacency matrix of both Gs and we imported their degrees into another matrix following the formula provided in the course we should get the final and correct eigenvector for both done as follows **screenshot**

- Results:

After doing all of this we can now comfortably export our results from R as follows **screenshot**

Which will export to us these files (after converting them to excel files) **screenshot**

Let's talk briefly about what each of the centralities mean given we are so close to us showcasing our results

1. **Degree** centrality is about how many direct connections a node has with other nodes meaning a node becomes more important based on how many connections it has, the greater the better!
2. **Betweenness** centrality shows us when trying to reach a node from another node, how frequent some nodes will be on that path, the more they show up in the path means any other nodes are more likely to use that path as will making that node more important
3. **Closeness** centrality shows us how much are some nodes are like bridges between clusters of nodes, meaning if a node has low closeness that would

- account into them being closer to most of the network parts making them a reliable path for the entire node
4. **Eigenvector** centrality is the importance of a node based on many a node has a connection with other high degree nodes spreading their influence more over that net

For the degree centrality (the higher the number the more important a node is) **screenshot**

For the betweenness centrality (the higher the number the more important a node is) **screenshot**

For the closeness centrality (the smaller the number the more important a node is) **screenshot**

For the eigenvector centrality (the higher the number the more important a node is) **screenshot**

- **Conclusion:** these results do show us that the node(protein) TP53 is one of the most important proteins in our body it having a high centrality score means its plays a very big role in our protein network which in turn helps us fight off a multitude of viruses and diseases

-Finally

Team credits

The students who worked in this project are:

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